z-score for a single value
If $X\sim N(\mu,\sigma^2)$, then for a single observation x,
$$z=\frac{x-\mu}{\sigma}$$

Ch8: Sampling Distributions

Central Limit Theorem
$X_1,\ldots,X_n$ independent from same dist. w/ mean μ and var. σ^2, good for $n\geq 30$ (as $n\to\infty$),
$$\bar{X}\approx N\left(\mu,\frac{\sigma^2}{n}\right), \qquad Z=\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\approx N(0,1)$$

Sample std. dev. $s=\sqrt{\dfrac{\sum_{i=1}^n (x_i-\bar{x})^2}{n-1}}$

Theorem:
if $X_1,\ldots,X_n \overset{\text{iid}}{\sim} N(\mu,\sigma^2)$, then
$$\bar{X}\sim N\left(\mu,\frac{\sigma^2}{n}\right), \qquad \sum_{i=1}^n X_i\sim N(n\mu,n\sigma^2)$$

Sampling dist. of s^2
if s^2 is var. of r.s. of size n taken from normal dist.,
$$\frac{(n-1)s^2}{\sigma^2}=\frac{\sum_{i=1}^n (X_i-\bar{X})^2}{\sigma^2}\sim\chi^2_{n-1}$$
$E(s^2)=\sigma^2$, $\quad \text{Var}(s^2)=\frac{2\sigma^4}{n-1}$

Sampling dist. of diff. between two means
ind. samples of size n_1,n_2 are drawn from two pop. with means μ_1,μ_2 and var. σ_1^2,σ_2^2, and n_1,n_2 are large,
$$\bar{X}_1-\bar{X}_2\approx N\left(\mu_1-\mu_2,\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}\right)$$

Law of Large Numbers
$X_1,X_2,\ldots$ w/ mean μ and var σ^2
$\bar{X}_n=\frac{1}{n}\sum_{i=1}^n X_i$ as $n\to\infty$
$\text{Var}(\bar{X}_n)=\frac{\sigma^2}{n}\to 0$
$E(\bar{X}_n)=\mu$ so $\bar{X}_n\to\mu$

Chebyshev's inequality
$$P(\mu-k\sigma<X<\mu+k\sigma)\geq 1-\frac{1}{k^2}, \qquad k>1$$

Markov's inequality
If $X\geq 0$ and $a>0$, then $P(X\geq a)\leq\dfrac{E(X)}{a}$

F-Distribution
If $U\sim\chi^2(\nu_1)$ and $V\sim\chi^2(\nu_2)$ are independent, then
$$F=\frac{U/\nu_1}{V/\nu_2}\sim F(\nu_1,\nu_2), \qquad f>0$$
right-skewed

Ch9: One and Two Sample Estimation

Common test statistics
$$z=\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}, t=\frac{\bar{x}-\mu}{s/\sqrt{n}}, \chi^2=\frac{(n-1)s^2}{\sigma^2}, F=\frac{s_1^2}{s_2^2}$$

Point estimate: single value $\hat{\theta}$ (e.g. $\bar{x}$) estimating population parameter θ (e.g. μ) of a statistic Θ (e.g. X)

Unbiased estimator: $E(\hat{\theta})=\theta$
(i.e. $E=$ true parameter)

Thrm: $E(\bar{X})=\mu$
Proof: $\bar{X}=\frac{1}{n}\sum_{i=1}^n X_i$
$\Rightarrow \quad E(\bar{X})=\frac{1}{n}\sum_{i=1}^n E(X_i)=\frac{n\mu}{n}=\mu$

Thrm: $E(s^2)=\sigma^2$
Proof: $\sum_{i=1}^n(X_i-\bar{X})^2=\sum_{i=1}^n\left[(X_i-\mu)-(\bar{X}-\mu)\right]^2$
$=\sum_{i=1}^n(X_i-\mu)^2-n(\bar{X}-\mu)^2$
$E(s^2)=E\left[\frac{1}{n-1}\sum(X_i-\bar{X})^2\right]$
$=\frac{1}{n-1}\left[\sum E[(X_i-\mu)^2]-nE[(\bar{X}-\mu)^2]\right]$
$=\frac{1}{n-1}\left[n\sigma^2-n\frac{\sigma^2}{n}\right]=\sigma^2$

Efficient estimator: smallest variance among unbiased estimators - $\text{Var}(\hat{\theta})$ is minimal

Std. error: std. dev. of sampling dist.
If $X_i\sim N(\mu,\sigma^2)$, $\bar{X}\sim N\left(\mu,\frac{\sigma^2}{n}\right)$ $\to$SD$=\sigma$, SE$=\frac{\sigma}{\sqrt{n}}$

CI for μ (σ known) $\bar{X}\sim N\left(\mu,\frac{\sigma^2}{n}\right)\Rightarrow$ if sample is from normal or by CLT
$P(L<\mu<U)=1-\alpha$
$$P\left(-z_{\alpha/2}<\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}<z_{\alpha/2}\right)=1-\alpha$$
$$P\left(\bar{X}-z_{\alpha/2}\frac{\sigma}{\sqrt{n}}<\mu<\bar{X}+z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)=1-\alpha$$

Sample size for desired error
$100(1-\alpha)\%$ confident that $|\bar{X}-\mu|$ will not exceed $z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
$$|\bar{X}-\mu|\leq z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\leq e, \qquad n\geq\left(\frac{z_{\alpha/2}\sigma}{e}\right)^2$$

CI for μ (σ unknown)
sample from $N(\mu,\sigma^2)$
$(1-\alpha)100\%$ CI: $\quad \bar{X}\pm t_{\alpha/2,\,n-1}\frac{S}{\sqrt{n}}$

Prediction Interval (PI)
interval for a single new observation X_0
$\bar{X}\sim N\left(\mu,\frac{\sigma^2}{n}\right)$, $\quad X_0\sim N(\mu,\sigma^2)$ (indep.)
$$X_0-\bar{X}\sim N\left(0,\sigma^2+\frac{\sigma^2}{n}\right)$$
(1) σ known: $\quad \bar{X}\pm z_{\alpha/2}\,\sigma\sqrt{1+\frac{1}{n}}$
(2) σ unknown: $\quad \bar{X}\pm t_{\alpha/2,\,n-1}\,s\sqrt{1+\frac{1}{n}}$

Two samples from normal
(1) Variances known
$$\bar{X}_1-\bar{X}_2\sim N\left(\mu_1-\mu_2,\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}\right)$$
$$P\left(-z_{\alpha/2}<\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{\sigma_1^2/n_1+\sigma_2^2/n_2}}<z_{\alpha/2}\right)=1-\alpha$$
CI: $\quad \bar{X}_1-\bar{X}_2\pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}$
(2) Variances unknown but equal
$\sigma_1^2=\sigma_2^2=\sigma^2$
$$\bar{X}_1-\bar{X}_2\sim N\left(\mu_1-\mu_2,\sigma^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)\right)$$
$$\hat{\sigma}^2=s_p^2=\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$$ **(pooled var.)**
CI: $\bar{X}_1-\bar{X}_2\pm t_{\alpha/2,n_1+n_2-2}s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$
careful: s_p, **not** s_p^2
(3) Variances unknown & unequal
$$T=\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{s_1^2/n_1+s_2^2/n_2}}\sim t_\nu$$
$$\nu=\frac{\left(s_1^2/n_1+s_2^2/n_2\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1-1}+\frac{\left(s_2^2/n_2\right)^2}{n_2-1}}$$ (round down to int.)
CI: $\quad \bar{X}_1-\bar{X}_2\pm t_{\alpha/2,\,\nu}\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}$

CI for single variance σ^2
If sample is from a normal population,
$$\text{CI:}\quad \frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}}<\sigma^2<\frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}$$
For std. dev. take **sqrt of the interval** itself.

CI for ratio of two variances
If ind. samples are from normal pop.,
$$\text{CI:}\quad \frac{s_1^2}{s_2^2}\frac{1}{f_{\alpha/2}(\nu_1,\nu_2)}<\frac{\sigma_1^2}{\sigma_2^2}<\frac{s_1^2}{s_2^2}f_{\alpha/2}(\nu_1,\nu_2)$$
where $\nu_1=n_1-1$, $\nu_2=n_2-1$

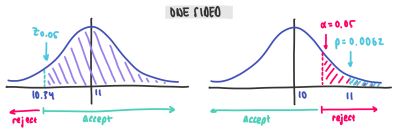
Paired observations: two obs. on same individual
$d_i=X_{1i}-X_{2i}$, $\quad i=1,\ldots,n$ *(n pairs, 2n obs.)*
$\text{Var}(d_i)=\text{Var}(X_{1i}-X_{2i})=\sigma_1^2+\sigma_2^2-2\,\text{Cov}(X_{1i},X_{2i})$
CI for $\mu_d=\mu_1-\mu_2$: $\bar{d}\pm t_{\alpha/2,n-1}\frac{s_d}{\sqrt{n}}$

CI for proportion p (large n)
X: number of successes in n trials. Let $\hat{p}=\frac{x}{n}$, $\quad \hat{q}=1-\hat{p}$.
$\hat{p}=\frac{x}{n}$, $\quad (1-\alpha)100\%$ CI: $\hat{p}\pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$
One-sided lower ($H_1:p>p_0$): $\hat{p}-z_\alpha\sqrt{\frac{\hat{p}\hat{q}}{n}}$
One-sided upper ($H_1:p<p_0$): $\hat{p}+z_\alpha\sqrt{\frac{\hat{p}\hat{q}}{n}}$

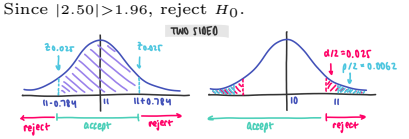
Maximum Likelihood Estimation (MLE)
Given a r.v. from a dist. w/ param. θ, the joint dist. is
$f(x_1,x_2,\ldots,x_n|\theta)=f(x_1|\theta)f(x_2|\theta)\cdots f(x_n|\theta)$
Define likelihood function
$L(\theta|x_1,\ldots,x_n)=f(x_1,\ldots,x_n|\theta)$
MLE $\hat{\theta}$ is the value of θ that maximizes L.
Take derivative of L (or $\ln L$ if easier) and set to 0

Ch10: Hypothesis Tests

Example: One sided test (known σ)
Given $\bar{x}=11$, $n=49$, $\sigma=2.8$ (population s.d.)
Assume: $H_0:\mu\leq 10$ $\qquad H_a:\mu>10$
(1) 95% **lower CI**
$$\bar{x}-z_{0.05}\frac{\sigma}{\sqrt{n}}=11-1.645\left(\tfrac{2.8}{7}\right)=10.34$$
CI is $(10.34,\infty)$. **Reject H_0 if $\mu_0\notin$ CI.**
$10\notin(10.34,\infty)$, so reject H_0.
(2) p-value: Assume H_0 true, so $\mu=10$.
$$z=\frac{11-10}{2.8/7}=2.50$$
$p=P(Z>2.50)=0.0062<0.05=\alpha\Rightarrow$reject H_0
Reject H_0 if $p<\alpha$.



Example: Same but two-sided (known σ)
Assume: $H_0:\mu=10$ $\qquad H_a:\mu\neq 10$
(1) 95% **CI**
$$\bar{x}\pm z_{0.025}\frac{\sigma}{\sqrt{n}}=11\pm 1.96\left(\tfrac{2.8}{7}\right)=11\pm 0.784$$
CI is $(10.216,11.784)$. $10\notin(10.216,11.784)$, so reject H_0.
(2) p-value: Assume H_0 true, so $\mu=10$.
$$z=\frac{11-10}{2.8/7}=2.50$$
$p=2P(Z>2.50)=2(0.0062)=0.0124<0.05=\alpha\Rightarrow$ reject H_0
(3) critical val: $z_{crit}=z_{\alpha/2}=z_{0.025}=1.96$
test statistic: $z=\frac{11-10}{2.8/7}=2.50$
Reject H_0 if $|z|>1.96$.
Since $|2.50|>1.96$, reject H_0.



Type 1: $\alpha=P(\text{reject } H_0|H_0 \text{ true})$
Type 2: $\beta=1-P(\text{reject } H_0|H_0 \text{ false})$ *[1−power]*

Example: two-sided t-test (unknown σ)
Given r.s of $n=20$ orders, $\bar{x}=11$, $s=2.8$, $\alpha=0.05$
$H_0:\mu=10$ $\qquad H_a:\mu\neq 10$
(1) Two-sided 95% **CI**
$$\bar{x}\pm t_{0.025,19}\frac{s}{\sqrt{n}}=11\pm 2.093\left(\frac{2.8}{\sqrt{20}}\right)\approx 11\pm 1.31$$
CI is $(9.69,12.31)$. $10\in$ CI $\Rightarrow$ fail to reject H_0.
(2) p-value
Assume H_0 true, $\mu=10$
$$t=\frac{11-10}{2.8/\sqrt{20}}\approx 1.60$$
$p=2P(T_{19}>1.60)\approx 0.13>0.05\Rightarrow$fail to reject H_0
(3) critical val: $t_{crit}=t_{\alpha/2,19}=t_{0.025,19}=2.093$
Test statistic: $t=\frac{11-10}{2.8/\sqrt{20}}\approx 1.60$
Reject H_0 if $|t|>2.093$.
Since $|1.60|<t_{crit}=2.093$, fail to reject H_0.

Exact binomial test for proportion p
Let $X=$ number of successes in sample size n.
Under $H_0:p=p_0$, $X\sim\text{Bin}(n,p_0)$.
left-tail: $H_1:p<p_0$: p-value$=P(X\leq x|p=p_0)$
right-tail: $H_1:p>p_0$: p-value$=P(X\geq x|p=p_0)$
two-tail: $H_1:p\neq p_0$:
if $x<np_0$, p-value$=2P(X\leq x|p=p_0)$;
if $x>np_0$, p-value$=2P(X\geq x|p=p_0)$;
Reject H_0 if p-value $\leq\alpha$.

Example: two-sided exact binomial test
Test $H_0:p=0.70, H_a:p\neq 0.70, \alpha=0.10$, Sample: $n=15$, $x=8$, so $\hat{p}=\frac{8}{15}$. Under H_0, $X\sim$b(x;15,0.70).
Since $x=8<np_0=15(0.70)=10.5$, p-value $=2P(X\leq 8)=2\sum_{k=0}^{8}\binom{15}{k}(0.7)^k(0.3)^{15-k}>0.10$, fail to reject H_0.

Two-proportion z-test
$H_0:p_1=p_2$, $\quad H_a:p_1\neq p_2$ (or $>,<$)
$\hat{p}_1=\frac{x_1}{n_1}$, $\quad \hat{p}_2=\frac{x_2}{n_2}$, $\quad \hat{p}=\frac{x_1+x_2}{n_1+n_2}$, $\quad \hat{q}=1-\hat{p}$
Test statistic: $Z=\dfrac{\hat{p}_1-\hat{p}_2}{\sqrt{\hat{p}\,\hat{q}((1/n_1)+(1/n_2))}}$
Reject H_0 if $|Z|>z_{\alpha/2}$ (two-sided)

Chi-Squared Goodness-of-Fit
H_0: data follow given dist., H_a: not follow
O_i: observed counts, E_i: expected counts
$$\chi^2=\sum_{i=1}^k\frac{(O_i-E_i)^2}{E_i}\sim\chi^2_{k-1}$$
Require: $E_i\geq 5$, reject if $\chi^2>\chi^2_{\alpha,\,k-1}$

F Test for Two Variances
If S_1^2,S_2^2 are sample variances from independent normal populations, then
$$F=\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\sim F(n_1-1,n_2-1)$$
If $H_0:\sigma_1^2=\sigma_2^2$, then
$$F=\frac{S_1^2}{S_2^2}\sim F(n_1-1,n_2-1)$$
Two sided, $s_1^2>s_2^2$: Reject if $F>f_{\alpha/2}$.
One-sided, $s_1^2>s_2^2$: Reject if $F>f_\alpha$.
Thrm: $f_{1-\alpha}(\nu_1,\nu_2)=\frac{1}{f_\alpha(\nu_2,\nu_1)}$

Ch11: Simple Linear Regression

Simple linear regression
$Y=\beta_0+\beta_1 x+\varepsilon$, $\quad E(Y)=\beta_0+\beta_1 x$
(1) Residuals are normally dist ($\varepsilon_i\overset{iid}{\sim}N(0,\sigma^2)$)
(2) Const. var. ($\text{Var}(Y)=\sigma^2$) (see res. spread)
(3) Linear rel. b/w mean response + predictor
(4) Errors are independent
β_0,β_1 : true (unknown) parameters
b_0,b_1 : estimates (computed from sample)
B_0,B_1 : random variables (estimators)

Residual: $e_i=y_i-\hat{y}_i$
$\text{SSE}=\sum e_i^2=\sum(y_i-\hat{y}_i)^2=\sum(y_i-b_0-b_1x_i)^2=$
$S_{yy}-b_1S_{xy}=S_{yy}-\dfrac{S_{xy}^2}{S_{xx}}$

LSE: Estimate β_0, β_1 by minimizing SSE
$\hat{y}=b_0+b_1x$, where $b_0=\hat{\beta}_0$, $b_1=\hat{\beta}_1$
$$\hat{\beta}_1=b_1=\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}=\frac{S_{xy}}{S_{xx}}$$
$\hat{\beta}_0=b_0=\bar{y}-b_1\bar{x}$

Properties of LSE
$E(B_1)=\beta_1$, $\quad E(B_0)=\beta_0$
$$B_1\sim N\left(\beta_1,\frac{\sigma^2}{S_{xx}}\right), \qquad B_0\sim N\left(\beta_0,\ \sigma^2\left(\frac{\Sigma x_i^2}{nS_{xx}}\right)\right)$$

MSE
$$s^2=\frac{\text{SSE}}{n-2}=\frac{\sum(y_i-\hat{y}_i)^2}{n-2}, \qquad E(s^2)=\sigma^2$$

CI for slope
CI of β_1: $b_1\pm t_{\alpha/2,\,n-2}\dfrac{s}{\sqrt{S_{xx}}}$
$S_{xx}=\sum(x_i-\bar{x})^2$, $\dfrac{s}{\sqrt{S_{xx}}}$ is the **standard error**.

Linear regression: test for slope $H_0:\beta_1=0$
$$t=\frac{b_1}{s/\sqrt{S_{xx}}}=\frac{\text{estimate}}{\text{std. error}}$$

CI for mean response at $x=x_0$
CI: $\hat{y}_0\pm t_{\alpha/2,\,n-2}\,s\sqrt{\dfrac{1}{n}+\dfrac{(x_0-\bar{x})^2}{S_{xx}}}$
where $\hat{y}_0=b_0+b_1x_0$

PI for new observation at $x=x_0$
PI: $\hat{y}_0\pm t_{\alpha/2,\,n-2}\,s\sqrt{1+\dfrac{1}{n}+\dfrac{(x_0-\bar{x})^2}{S_{xx}}}$

$$\underbrace{\sum_{i=1}^n(y_i-\bar{y})^2}_{\text{SST}}=\underbrace{\sum_{i=1}^n(\hat{y}_i-\bar{y})^2}_{\text{SSR}}+\underbrace{\sum_{i=1}^n(y_i-\hat{y}_i)^2}_{\text{SSE}}$$

Example: size and power for a two-sided z-test
$X_1,\ldots,X_{100}\sim N(\mu,25)$.
$H_0:\mu=10$, $H_a:\mu\neq 10$, reject H_0 if $\bar{X}\geq 10.98$ or $\bar{X}\leq 9.02$.
(1) Size α
Under H_0, $\mu=10$, so $\bar{X}\sim N(10,25/100)$.
$\alpha=P(\bar{X}\geq 10.98)+P(\bar{X}\leq 9.02)$
$$=2P\left(Z>\frac{10.98-\mu}{\sigma/\sqrt{n}}\right)=2P\left(Z>\frac{10.98-10}{0.5}\right)=0.05$$
(2) Power at $\mu=10.5$
Under $\mu=10.5$, $\bar{X}\sim N(10.5,25/100)$.
$\text{Power}=P(\bar{X}\geq 10.98)+P(\bar{X}\leq 9.02)$
$$=P\left(Z>\frac{10.98-10.5}{0.5}\right)+P\left(Z<\frac{9.02-10.5}{0.5}\right)$$
$=0.1685+0.0015=0.17$
Idea: for power, use the *true* parameter value in the distribution, then add the probabilities of falling in the rejection region.

residuals/qq: Plot is fairly rand. (no systematic pattern), stdized res. fall approx. straight line (some deviations = outliers; or all b/w [-x, x] = no potential outliers), thus it is normally dist. Assumptions seem to hold.
residuals: var. of res. inc. w/ regressor var. x (no const. variance). Departures from rand. scatter, need lin/quad term in regressor x

```
> anova(fit)
Analysis of Variance Table

Response: heat
          Df Sum Sq Mean Sq F value    Pr(>F)
work       1  94659   94659   181.1 1.404e-15 ***
Residuals 22   5232     238
---
> summary(fit)
Call:
lm(formula = heat ~ work, data = phys)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 126.5588     7.4789   16.92 3.36e-14 ***
work          3.9212     0.2914   13.45 1.40e-15 ***
---

Residual standard error: [F] on 22 degrees of freedom
Multiple R-squared: 0.9076, work explains 90% of the variance
```